MATH290

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Proposition 1. *For any integer k, $k^3 - k$ is a multiple of 6.*

Proof.

Firstly, recall that an integer is a multiple of an integer 6 if it can be expressed as the product of 6 and some other integer. Now, observe what happens if you substitute $-k$ for k:

$$\begin{aligned} &(-k)^3 - (-k) \\ =& -(k^3) + k \\ =& -(k^3 - k) \end{aligned}$$

Thus, if $k^3 - k$ can be expressed as $6n$ for some integers k and n, then $(-k)^3 - (-k)$ can be expressed as $6(-n)$. Therefore, if I can show that $k^3 - k$ is a multiple of 6 for all integers $k \geq 0$, then $k^3 - k$ is a multiple of 6 for all integers $k < 0$. With this in mind, I'll focus only on integers $k \geq 0$ for the rest of this proof.

Let f_k be the sequence $k^3 - k$ for integers $k \geq 0$. It turns out that rather than proving each term in this sequence is a multiple of 6, it's easier to prove that *the difference between consecutive terms* is a multiple of 6. (Or at least *I* found it to be.) Let's let g_k be the difference between any f_{k+1} and f_k. That is,

$$\begin{aligned} g_k &= f_{k+1} - f_k \\ f_{k+1} &= f_k + g_k \end{aligned}$$

If I can show that g_k is a multiple of 6 for all integers k, then so long as the *first* term of f is a multiple of 6, *all* terms of f are multiples of 6. This relies on the property that the sum of two multiples of 6 is itself a multiple of 6. I'll briefly demonstrate this property for two multiples of six $n = 6i$ and $m = 6j$, where n, m, i, and j are all integers. Observe,

$$\begin{aligned} n + m &= 6i + 6j \\ &= 6(i + j) \end{aligned}$$

Relying on the property that summing integers yields an integer, we've shown that $n + m$ can be expressed as a product of 6 and some other integer $i + j$, and thus $n + m$ is a multiple of 6.

Let's start by showing that g_k is a multiple of 6 for all integers k. Because 6 is equal to 2×3, I can prove g_k is a multiple of 6 by proving it is a multiple of both 2 and 3. I can prove g_k is always a multiple of 3 by showing it can be expressed as the product of 3 and some other integer. Observe,

$$\begin{aligned} g_k &= f_{k+1} - f_k \\ &= (k+1)^3 - (k+1) - (k^3 - k) \\ &= (k+1)\big((k+1)^2 - 1\big) - (k^3 - k) \\ &= (k+1)(k^2 + 2k) - (k^3 - k) \\ &= k^3 + 3k^2 + 2k - k^3 + k \\ &= 3k^2 + 3k \\ &= 3(k^2 + k) \end{aligned}$$

To prove that g_k is a multiple of 2 for all integers k, I will prove that g_k is *even* for all integers k. Firstly, recognize that for any integer k, if $k^2 + k$ is even, then $3(k^2 + k)$ is even. Then, recognize that for any *even* integer k, both k^2 and k are even, and so their sum $k^2 + k$ is even. Lastly, recognize that for any *odd* integer k, both k^2 and k are odd, and so $k^2 + k$ is even. So, regardless of whether an integer k is even or odd, $k^2 + k$ is even and therefore means $g_k = 3(k^2 - k)$ is also even.

(This line of reasoning relies on the knowledge that the product of an even and an odd is even; the square of an even is even; the square of an odd is odd; the sum (or difference) of two evens is even; and the sum (or difference) of two odds is even.)

Therefore, having shown that g_k is a multiple of 3 and even for every integer k, I have proven that g_k is a multiple of 6 for every integer k.

The rest can be proven by mathematical induction on k. Suppose that for some integer $k \geq 0$, f_k is a multiple of 6. Consider then the case for $k + 1$: $f_{k+1} = f_k + g_k$. Because f_k and g_k are both multiples of 6, their sum, f_{k+1}, is also a multiple of 6. Thus, the result holds for the inductive case.

Finally, consider the base case f_0. Observe,

$$\begin{aligned} f_0 &= (0)^3 - (0) \\ &= 0 \end{aligned}$$

0 is a multiple of 6 (it can be expressed as 6×0). Thus, the result holds for the base case $k = 0$. Then, by the principle of mathematical induction, $k^3 - k$ is a multiple of 6 for every integer $k \geq 0$. And, therefore, $k^3 - k$ is a multiple of 6 for every integer $k < 0$ also. Altogether, $k^3 - k$ is a multiple of 6 for every integer k.

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